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**YISHUN JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2011**

BIOLOGY

9648/01

HIGHER 2

25 AUG 2011

Paper 1 Multiple Choice

**Thurs 1400 – 1515
1 hour 15 minutes**

Additional material: Multiple Choice Answer Sheet



READ THESE INSTRUCTIONS FIRST

Write in soft pencil (type B or HB is recommended).

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your Centre number, index number, name and CTG on the Answer Sheet in the spaces provided unless this has been done for you.

There are **twenty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

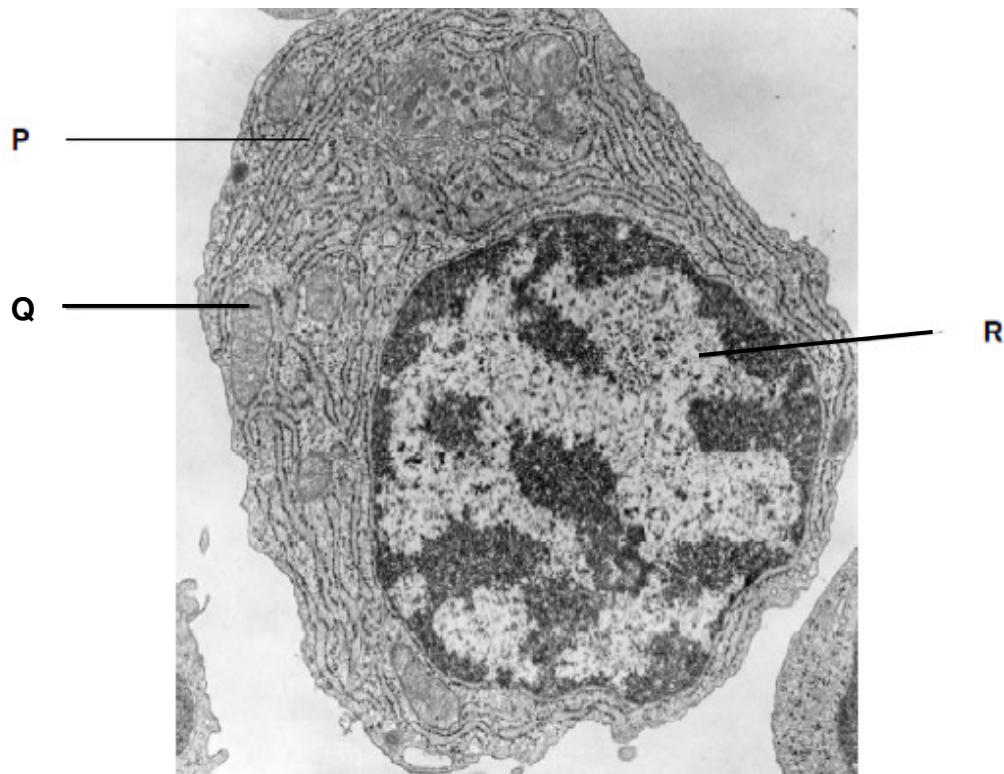
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet. Calculators may be used.

This question paper consists of **30** printed pages.

- 1 The figure shows an electron micrograph of a cell.



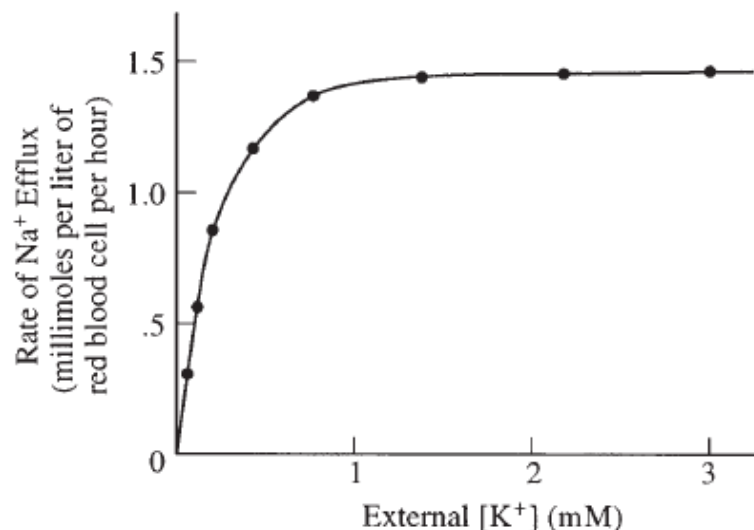
Which of the following about structures **P**, **Q** and **R** are correct?

	P	Q	R
A	Provides large surface area for attachment of ribosomes	Modification of transcripts	Active condensation of chromatin
B	Transport of proteins to Golgi apparatus	Formation of ATP from light energy	Active transcription of genes
C	Synthesis and processing of membrane proteins	Isolating of protons within a specialised compartment	Active transcription of genes
D	Prevents proteins from degradation by cytoplasmic enzymes	Allows the oxidation of pyruvate	Active replication of DNA

2 Why do erythrocytes swell and burst when placed in water?

- A** Since water potential is higher outside the cell, water moves inward by passive diffusion.
- B** Since haemoglobin concentration is higher inside the cell, haemoglobin moves outward by exocytosis.
- C** Since potassium ions are more concentrated inside the cells, potassium ions move outward by osmosis.
- D** Erythrocytes pump water inward by active transport to balance osmotic gradients.

3 A freshly isolated population of mature human red blood cells was rapidly loaded with radioactive sodium and then subdivided into equally sized samples. The samples were incubated in isosmotic media containing 120-millimolar sodium chloride and different concentrations of potassium chloride, under otherwise identical conditions. In the graph below, the rates at which sodium left the cells are plotted as a function of the extracellular potassium concentrations.

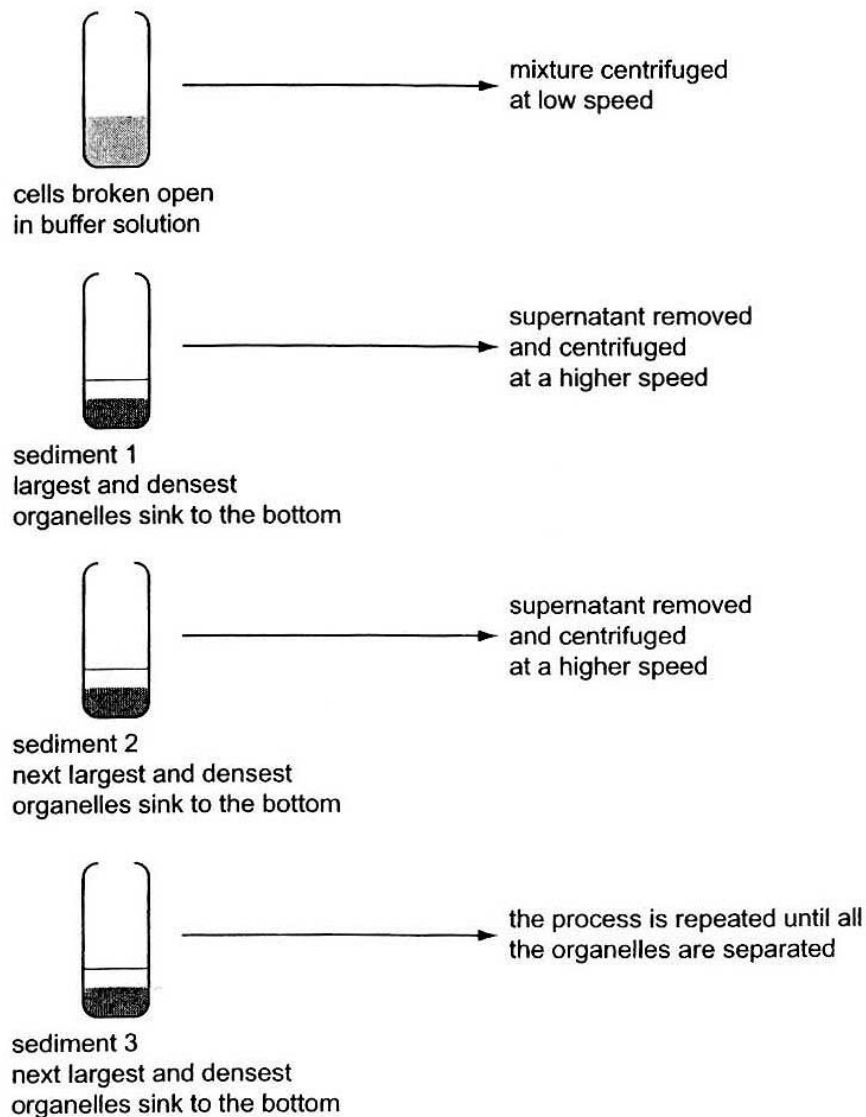


The rate of sodium efflux reaches a plateau at about 1.0 mM external potassium, because at that concentration, potassium

- A** competes with sodium for transport.
- B** occupies all the potassium-binding sites on the sodium carriers.
- C** makes the membrane impermeable to sodium.
- D** binds ATP, thereby lowering the substrate concentration.

- 4 Fractionation is a process used to separate cell components according to their size and density.

The diagram below shows the main steps in fractionation of a plant cell.

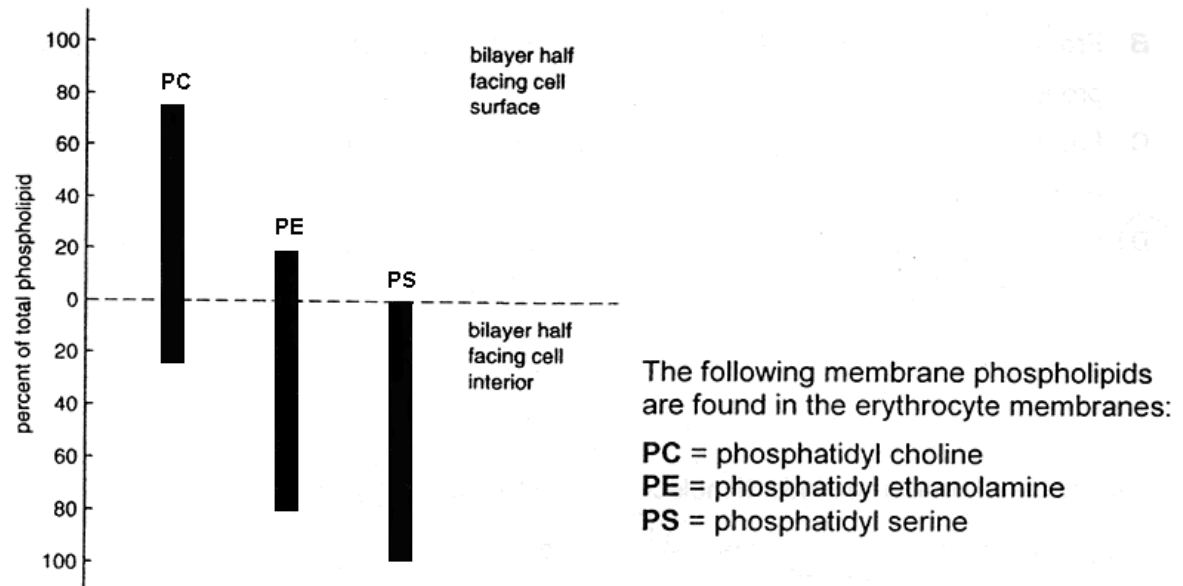


DCPIP and buffer solution were added to each sediment and the mixtures left in the light for fifteen minutes. Sediment 2 caused the oxidised DCPIP to be reduced.

Which organelles are present in sediment 2?

- A chloroplasts
- B Mitochondria
- C Nuclei
- D Ribosomes

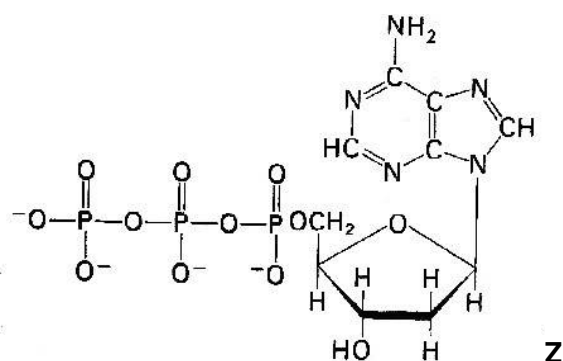
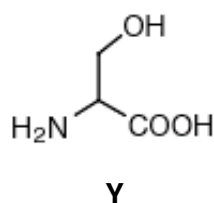
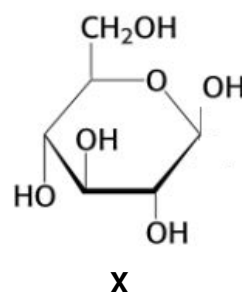
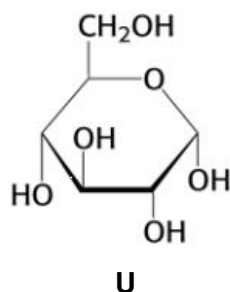
- 5 The following diagram illustrates the types of membrane phospholipids found in the intact erythrocytes (red blood cells).



Which of the following statements correctly identifies with the graph?

- I. 76% of the total membrane phospholipids contain choline and 20% contain ethanolamine.
 - II. 24% of **PC** and 80% of **PE** make up part of the inner membrane phospholipids.
 - III. Most **PC** is confined to the outer surface of the erythrocytes while most of the **PE** and **PS** are confined to the inner surface of the erythrocytes.
 - IV. Membranes, in general, can be concluded to be asymmetric.
- A** I and II only
B II and III only
C III and IV only
D II and IV only

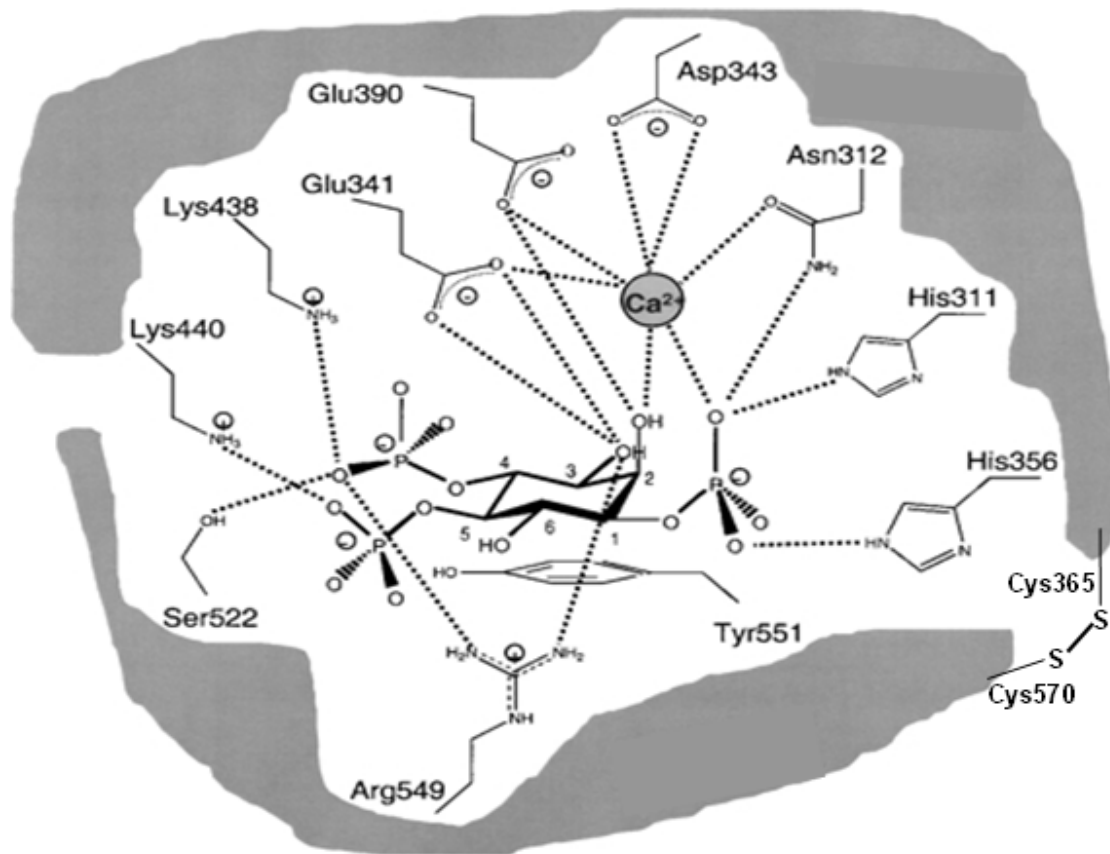
6 The diagram below shows the structure of 4 monomers.



Which of the following combination of polymer, monomer and bond formed between monomers is correct?

	<i>starch</i>	<i>cellulose</i>	<i>polypeptide</i>	<i>polynucleotide</i>
A	X , β -1,4 glycosidic bond	U , α -1,4 glycosidic bond	Z , ester linkage	Y , disulphide linkage
B	U , α -1,4 glycosidic bond	X , β -1,4 glycosidic bond	Y , peptide bond	Z , phosphoester linkage
C	Z , peptide bond	X , hydrogen bond	Z , ionic bond	U , hydrogen bond
D	X , ionic bonds	Y , peptide bond	U , hydrogen bond	Z , α -1,6 glycosidic bond

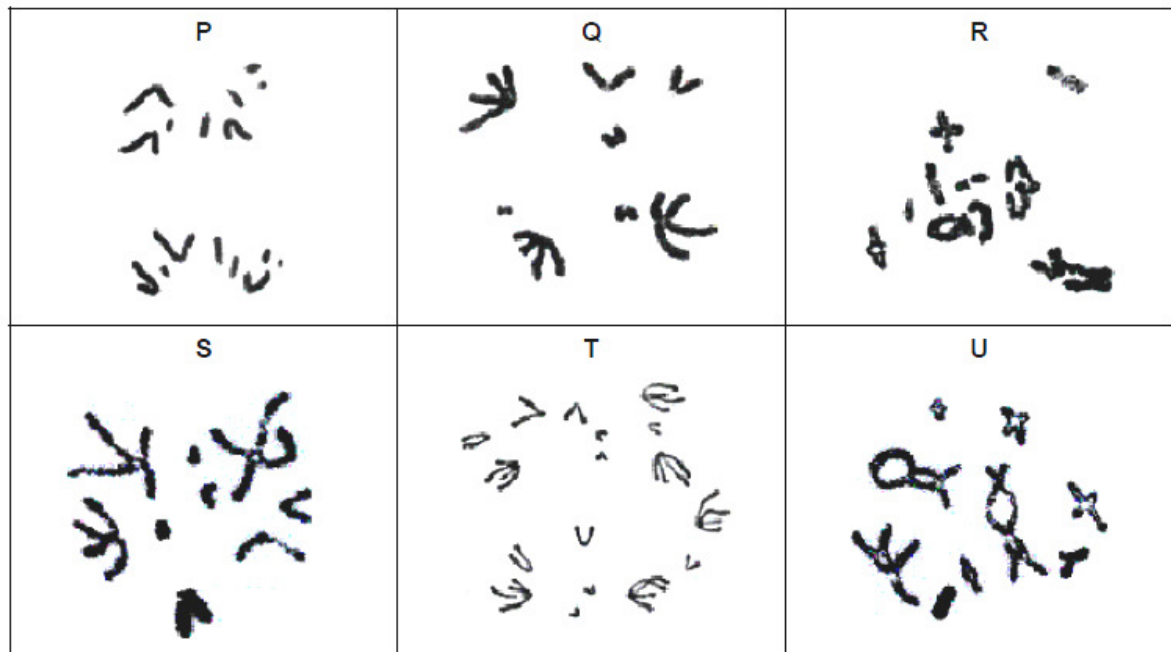
- 7 The diagram shows the interactions between the active site of phospholipase and its substrate. Of all the amino acids shown, only His311 and His356 actually take part in the reaction.



What are the roles of the following amino acids in phospholipase?

	His356	Tyr551	Cys570
A	contact residue	non-essential residue	structural residue
B	catalytic residue	contact residue	non-essential residue
C	catalytic residue	contact residue	structural residue
D	structural residue	non-essential residue	catalytic residue

- 8 The diagram below shows some of the events that take place during meiosis.



Which of the following shows the correct sequence of events?

- A $U \rightarrow T \rightarrow R \rightarrow S \rightarrow Q \rightarrow P$
- B $U \rightarrow R \rightarrow T \rightarrow S \rightarrow Q \rightarrow P$
- C $R \rightarrow Q \rightarrow U \rightarrow S \rightarrow P \rightarrow T$
- D $R \rightarrow U \rightarrow Q \rightarrow S \rightarrow T \rightarrow P$

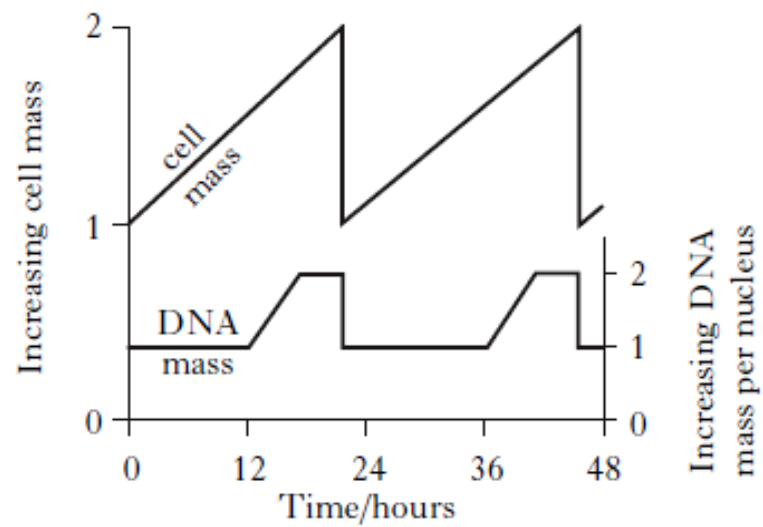
- 9 Lymphocytes extracted from the rat kangaroo, *Potorous tridactylus*, were grown in culture for karyotype analysis. To obtain chromosome spreads, such as that illustrated below, lymphocytes were incubated in the anti-mitotic drug, colchicines, for 30 minutes before fixation and staining. The chromosomes depicted below are representative of the karyotype found in all of the cells of the culture.



During anaphase, how many kinetochores would be expected in a normal rat kangaroo lymphocyte?

- A 12
 - B 18
 - C 24
 - D 48
- 10 Which of the following is LEAST likely to cause a proto-oncogene to become an oncogene?
- A A gene is incorporated into a retroviral genome.
 - B A gene is moved close to an enhancer, causing excess product to be made.
 - C A gene is truncated, yielding a protein with modified activity.
 - D A gene is moved into centromeric heterochromatin, silencing its transcription.

- 11 The diagram below shows the changes in cell mass and DNA mass during two cell cycles.

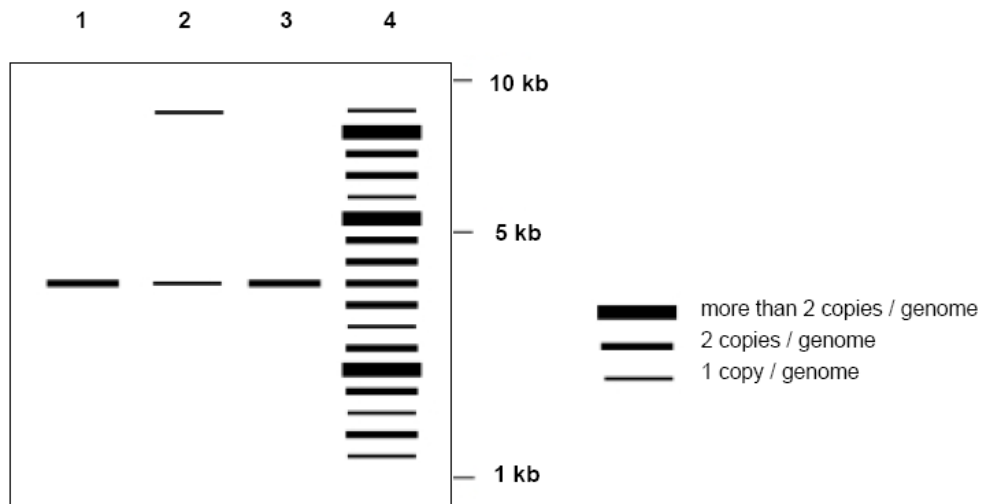


What valid conclusion could be made from the graph?

During the cell cycle,

- A interphase is the longest phase.
- B mitosis is divided into four phases.
- C replication takes place between 0 and 12 hours.
- D cytokinesis takes place at 12 and 36 hours.

- 12 The *ras* oncogene is involved in a variety of human cancers. DNA isolated from a number of normal and cancerous human tissues were digested with *Eco*RI. A Southern blot was performed and the membrane was probed with radioactive-labelled cloned *ras* DNA. After exposure to x-ray film, the resulting autoradiograph is shown below.



Lane	Sample
1	White blood cells from a healthy human
2	Lymphoma cells (cancerous)
3	Bladder carcinoma cells (cancerous)
4	Sarcoma cells (cancerous)

Based on the experimental results, which of the following statement(s) could be deduced?

I	In tissue 3, there was no mutation within the <i>ras</i> coding region.
II	The result for tissue 2 could be explained by separate point mutations that caused the removal of an <i>Eco</i> RI site and the creation of another within the probe DNA.
III	Gene amplification was involved in tissue 4.
IV	In tissue 3, the mutation did not affect existing <i>Eco</i> RI sites within the <i>ras</i> gene.

- A II only
- B I, II and IV
- C II, III and IV
- D III and IV

- 13 Cells of the bacterium *E. coli* were grown for many generations on a medium containing only the heavy isotope of nitrogen, ^{15}N . The cells were then transferred to a medium containing only ^{14}N and allowed to grow. Samples of the bacteria were removed from the culture after one generation. The DNA from each sample was extracted and centrifuged at very high speeds for several days in a solution of caesium chloride.



The diagram above illustrates the DNA distribution in various centrifuge tubes, **I**, **II**, **III**, **IV** and **V** as proposed by a student.

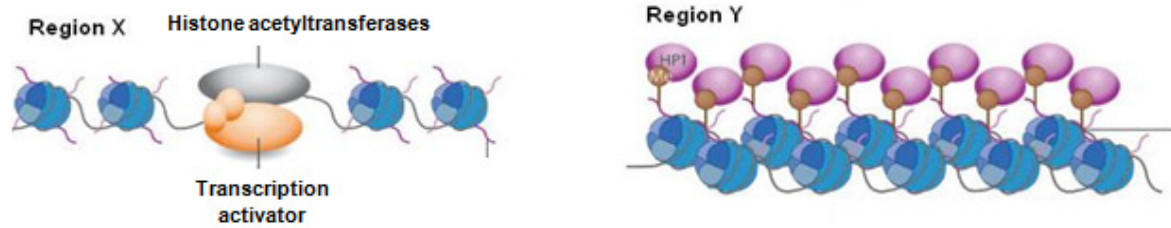
Which of the above centrifuge tubes would support the hypothesis of conservative DNA replication and which would support the hypothesis of semi-conservative DNA replication?

	conservative replication	semi-conservative replication
A	IV	II
B	II	III
C	III	I
D	V	I

- 14 What is the minimum number of GTP and ATP used in the synthesis of a pentapeptide?

	GTP	ATP
A	6	0
B	11	5
C	0	16
D	6	5

- 15 The diagrams show two different **regions X** and **Y** of a eukaryotic chromosome.



Regions X and **Y** are compared as follows:

	Region X	Region Y
I	Less condensed	Highly condensed
II	Contains unique sequence	Contains repetitive sequences
III	Gene-rich	Gene-poor
IV	Replicated during cell cycle	Not replicated during cell cycle
V	Recombination during meiosis	No recombination during meiosis

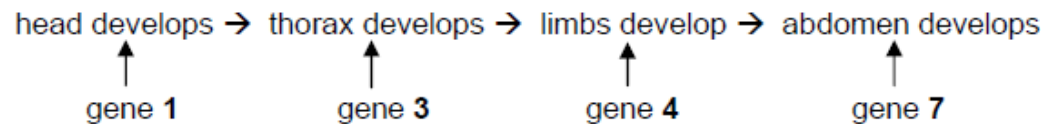
Which of the above comparisons are correct?

- A** I, III and IV only
- B** II, IV and V only
- C** I, II, III and V only
- D** All of the above

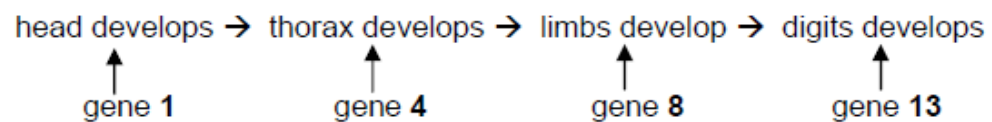
- 16** The normal order of development of features in an embryo is determined by gene sequences. The relevant groups of genes have been studied extensively in insects and chordates. These groups of genes are responsible for directing the sequential embryonic development of the head, thorax, limbs and wings of insects and the head, thorax and limbs in chordates.

In the summary below, gene activity starts at the left and continues in sequence to the right. The action of each gene in the group promotes the action of the subsequent gene.

In insects :



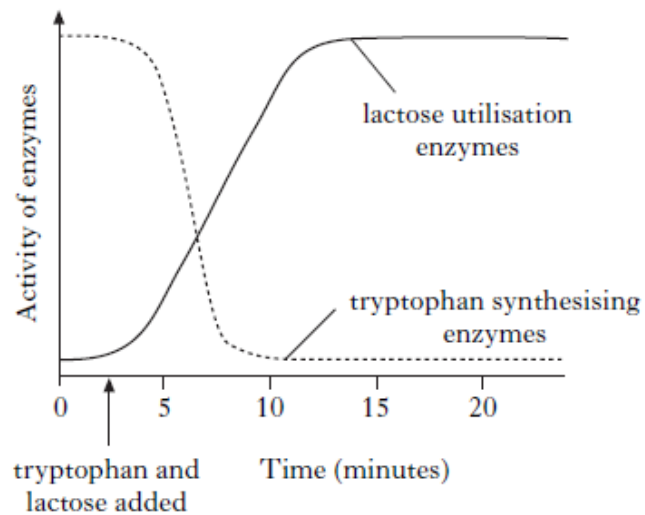
In vertebrates :



The above information suggests that if the action of a particular gene in a sequence was blocked,

- A** an abnormal insect could develop possessing a head but no legs.
- B** an abnormal insect could develop possessing a thorax and legs but no head.
- C** an abnormal chordate could develop with a thorax but no head.
- D** an abnormal chordate could develop possessing digits but no legs.

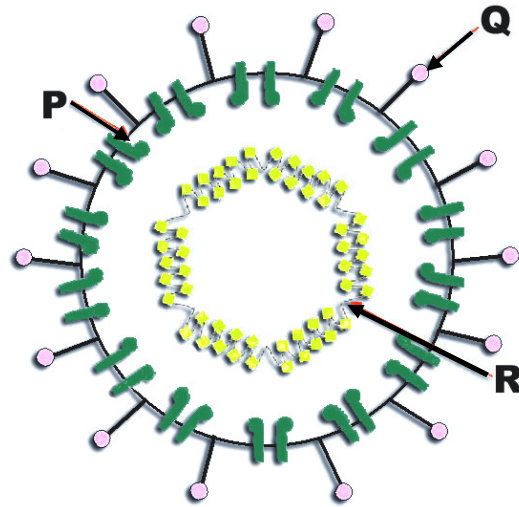
- 17 The diagram below shows the changes in the activity of enzymes that synthesise tryptophan and utilise lactose in a cell after the addition of tryptophan and lactose.



What valid conclusion may be made from the graph?

- A Addition of lactose acts as a negative enzyme modulator.
- B Addition of tryptophan acts as a positive enzyme modulator.
- C Enzyme induction is occurring in lactose utilisation enzymes.
- D Enzyme induction is occurring in tryptophan synthesising enzymes.

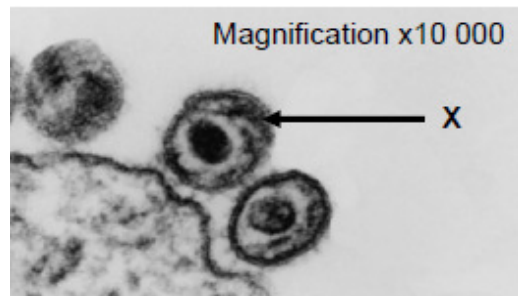
- 18 The diagram shows the structure of a severe acute respiratory syndrome coronavirus (SARS-CoV).



What are the labelled parts?

	glycoprotein	membrane	ribonucleic acid
A	P	Q	R
B	Q	P	R
C	R	P	Q
D	Q	R	P

- 19 The diagram below shows an electron micrograph of HIV particles leaving a cell of an AIDS patient.



Which of following statements are false?

- I. The size of HIV particle **X** is approximately 1.25×10^{12} nm.
 - II. The HIV particles recognise the host cell through the sialic-acid containing proteins or lipids on the membrane of the host cell.
 - II. Upon entry of the HIV into the host cell, a drop in pH will result in a conformational change in the HIV structure, consequently releasing the viral genome into the host cell.
 - IV. The viral DNA which enters the host cell's nucleus will be integrated into the genetic material of the host cell using the host cell's enzyme, integrase.
- A** II and IV only
- B** I and III only
- C** I, II and III only
- D** All of the above
- 20 In *Drosophila melanogaster*, cinnabar eye (*cn*) and vestigial wing (*vg*) are simple recessive traits. A female, heterozygous for both genes, was crossed with a male with cinnabar eyes and vestigial wings. The offspring resulting from this cross are listed in the table below.

Class	Phenotype	Number
I	Wild type	445
II	Cinnabar eye, wild-type wing	51
III	Wild-type eye, vestigial wing	49
IV	Cinnabar eye, vestigial wing	455

From the data shown, the distance between the *cn* and *vg* genes is approximately

- A** 500 map units.
- B** 100 map units.
- C** 50 map units.
- D** 10 map units.

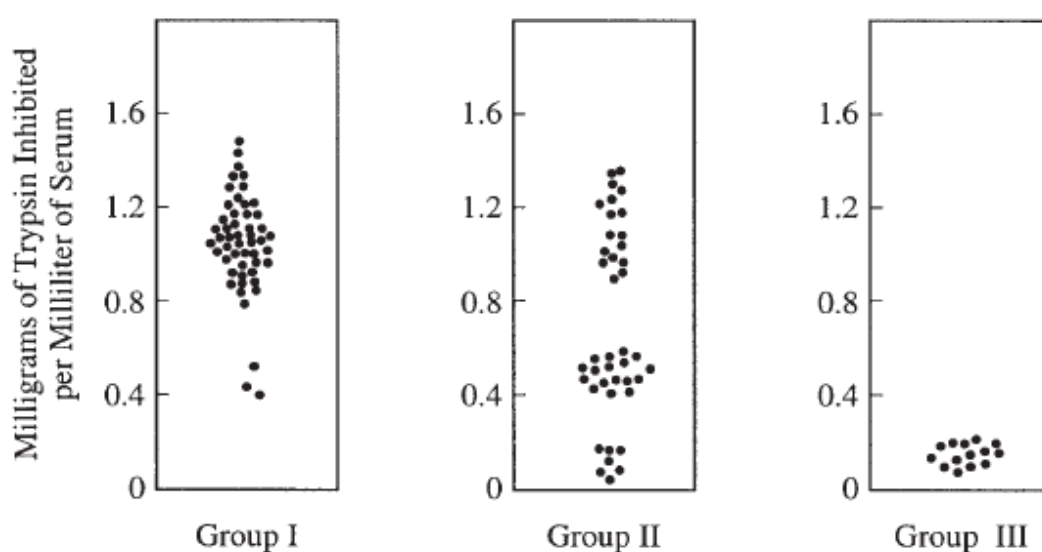
- 21 Some individuals suffer from an autosomal recessive disorder known as 'alpha₁-antitrypsin deficiency'. The recessive homozygote for this disorder lacks the enzyme that ordinarily degrades trypsin. The genotype of the recessive disorder may be designated as *aa*. Normal individuals are either homozygous dominant (*AA*) or heterozygous carriers (*Aa*).

Enzymatic tests and colorimetric analysis reveal that a given individual possesses one of the three distinct levels of alpha₁-antitrypsin enzyme activity. The data below show the levels of trypsin inhibited per millilitre of serum (which contains the antitrypsin enzyme) in three different groups.

Group I: The general population

Group II: Several families in which some members have alpha₁-antitrypsin deficiency

Group III: Patients with alpha₁-antitrypsin deficiency



When the serum of an adult woman was tested colorimetrically, a level of 0.5 milligram of trypsin inhibited per millimetre of serum was found. This can be interpreted as

- A She is homozygous dominant.
- B She is an asymptomatic heterozygous carrier.
- C She will exhibit alpha₁-antitrypsin deficiency.
- D Her mother is most likely homozygous dominant.

- 22 Two unlinked loci affect mouse fur colour. **CC** or **Cc** mice are agouti. Mice with genotype **cc** are albino because all pigment production and deposition of pigment in hair is blocked. At the second locus, the **B** allele (black agouti coat) is dominant to the **b** allele (brown agouti coat).

A mouse with a black agouti coat is mated with an albino mouse of genotype **bbcc**. Half of the offspring are albino, one quarter are black agouti, and one quarter are brown agouti.

What is the genotype of the black agouti parent?

- A **BbCc**
- B **BbCC**
- C **BBCc**
- D **BBCC**

Questions 23 and 24 are related to the information provided below.

In experiments to determine possible metabolic side effects of a new industrial by-product X941, the rate of respiratory oxygen consumption by isolated mitochondria was assayed. The mitochondria were maintained in a solution containing high concentrations of dissolved oxygen and inorganic phosphate.

Figure 1 shows the results of a control experiment. Pyruvic acid, the principal metabolic precursor for the citric acid cycle, was added to isolated mitochondria, followed a short time later by the addition of ADP. Figure 2 illustrates the effect of adding X941 instead of ADP.

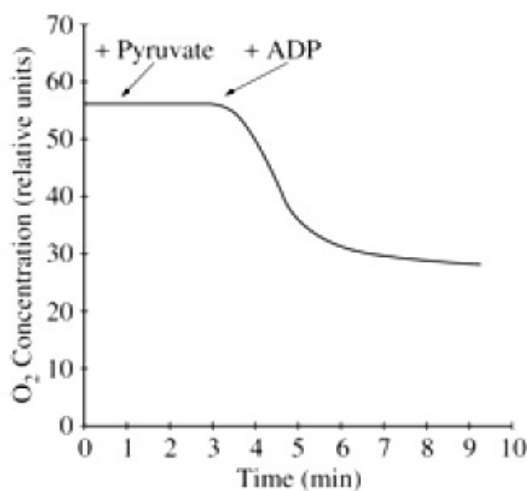


Figure 1

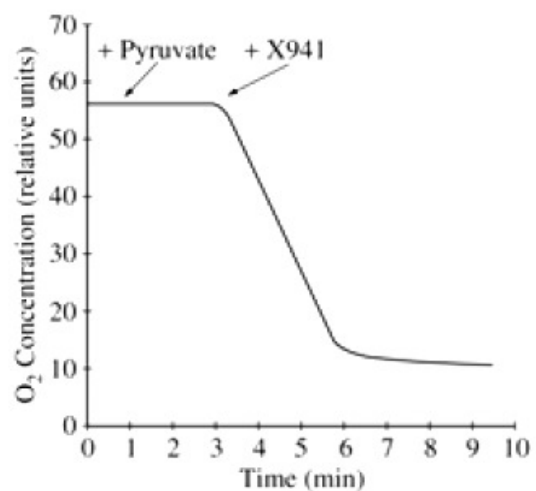


Figure 2

- 23** Which of the following best explains why O_2 concentration does not decrease until after the addition of ADP, as shown in Figure 1?
- A** The proton gradient that drives electron transport can be produced only by phosphorylation of ADP.
 - B** ADP is a substrate, along with pyruvate, for pyruvate dehydrogenase.
 - C** ADP is an allosteric activator of cytochrome oxidase, which is unable to transfer electrons to O_2 without ADP.
 - D** The proton gradient produced by pyruvate metabolism quickly inhibits electron transport; the inhibition is released by ATP synthesis.
- 24** In Figure 2, the addition of X941 results in the nearly complete depletion of the O_2 in the assay mixture. A further test showed that this depletion occurred even when inorganic phosphate was omitted from the assay. The most likely explanation for the effect of X941 is that X941
- A** allows O_2 to leak out of the mitochondria.
 - B** is a precursor to the synthesis of ADP.
 - C** uncouples ADP phosphorylation from electron transport.
 - D** serves as an alternative electron acceptor in the system.

Questions 25 and 26 are related to the information provided below.

During pregnancy, glucose is transferred from the bloodstream of the mother to the bloodstream of the foetus through the placenta, making use of hexose transporters that are not dependent on insulin.

In an experiment conducted on a pregnant female subject, experimental steps **X** and **Y** were done with control periods of no treatment between them.

Measurements of blood glucose levels in both mother and foetus were made. Also, the glucose transfer rates from mother to placenta, and from placenta to foetus were monitored. The experimental data is shown in the table below.

Experimental step	Glucose concentration / mg cm^{-3}		Glucose transfer rate / mg minute^{-1}	
	Maternal blood	Foetal blood	From mother to placenta	From placental to foetus
Control period	54	15	38	9
After X	54	9	38	16
Control period	52	14	39	8
After Y	211	30	58	34

- 25 Which of the following is likely to describe experimental steps **X** and **Y**?

	X	Y
A	Glucagon injection given to foetus	Insulin injection given to mother
B	Insulin injection given to foetus	Glucagon injection given to foetus
C	Insulin injection given to mother	Glucagon injection given to foetus
D	Insulin injection given to foetus	Glucagon injection given to mother

- 26 There was an approximately four-fold increase in maternal blood glucose after treatment **Y**. However, the rate of glucose transfer from mother to placenta only increased less than two-fold. Which of the following is the most likely explanation for this?

- A** Glucose transfer from mother to placenta occurs by facilitated diffusion.
- B** Glucose transport needs to occur down a glucose concentration gradient.
- C** There was an increase in hexose transporters for transferring glucose from mother to placenta.
- D** There is saturation of the number of available hexose transporters.

- 27 A common neurotransmitter is acetylcholine (ACh), which excites the post-synaptic neuron. The lack of ACh in brain synapses has been linked to Alzheimer's disease. A number of other chemicals also act as synapses. The table below outlines some of these chemicals and their actions.

Chemical	Source in nature	Effect at synapse
Botulinum	<i>Clostridium</i> bacteria	Prevents release of ACh from the pre-synaptic membrane
Esterine	Africa calabar bean	Inhibits acetylcholinesterase, the enzyme that breaks down ACh after post-synaptic transmission
Curare	South American plant <i>Chodrodendron</i> sp.	Blocks post-synaptic receptors
Nicotine	Tobacco plant	Stimulates in the same way as ACh

Based on this information, a possible treatment for Alzheimer's disease is to

- A** administer Esterine as a drug.
- B** inject curare into the patient's CNS.
- C** recommend that the patient stop smoking.
- D** purposely infect the patient with *Clostridium*.

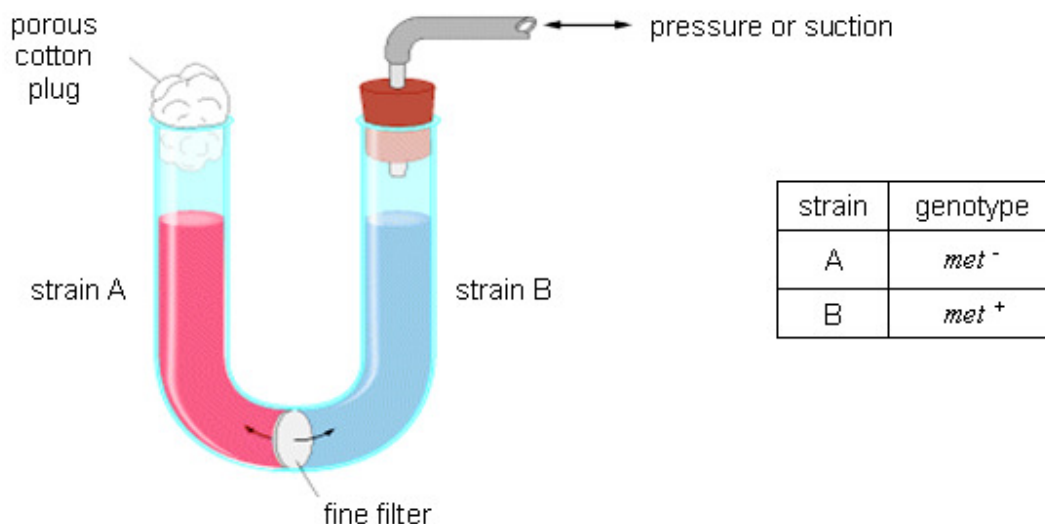
Questions 28 and 29 are related to the information provided below.

In order to study the activity of enzyme **X**, cells were treated with either glucagon, dibutyl cAMP, or glucagon plus H-8 at doses required to achieve a maximal effect. The activity of enzyme **X** was then assayed in dilute cell-free extracts. Dibutyl cAMP is an analog of cAMP that diffuses across cell membranes more easily than does cAMP itself. H-8 is a selective inhibitor of protein kinase A. The results are shown in the following table.

Agent added	Enzyme activity (units)	Enzyme concentration (micrograms per mL)
None	10	10
Glucagon	100	12
Dibutyl cAMP	100	9
Glucagon plus H-8	18	11

- 28** It can be inferred from these results that glucagon most likely acts to
- A** cause translocation of enzyme **X** to lysosomes.
 - B** increase the catalytic efficiency of enzyme **X**.
 - C** stimulate transcription of the gene for enzyme **X**.
 - D** stimulate increased intracellular production of the substrate for enzyme **X**.
- 29** It can be inferred from these results that changes in activity of enzyme **X** are mediated by
- A** a phosphorylation, probably catalysed by protein kinase A.
 - B** proteolytic cleavage of an intermediary protein.
 - C** a dephosphorylation, probably catalysed by protein phosphatase 1A.
 - D** a cAMP-independent signalling pathway.

- 30 To investigate gene transfer between bacteria, two strains of the same bacterial species were each placed in one arm of a U-tube with a filter separating them.



met⁺ is a wild-type gene that codes for the ability to synthesise the essential amino acid, methionine.

met⁻ indicates that the *met*⁺ gene has been mutated.

Liquid may be transferred between the arms of the tube by the application of pressure or suction, but particles that are larger than the filter pore size would not be able to pass through the fine filter.

type of particle	size
bacteria	1 – 10µm
bacteriophages	0.025 – 0.2µm

After several hours of incubation, bacterial cells from the left arm of the tube are plated on minimal medium.

Which pair of experimental results shows that transduction was most likely the process responsible for gene transfer between strains A and B?

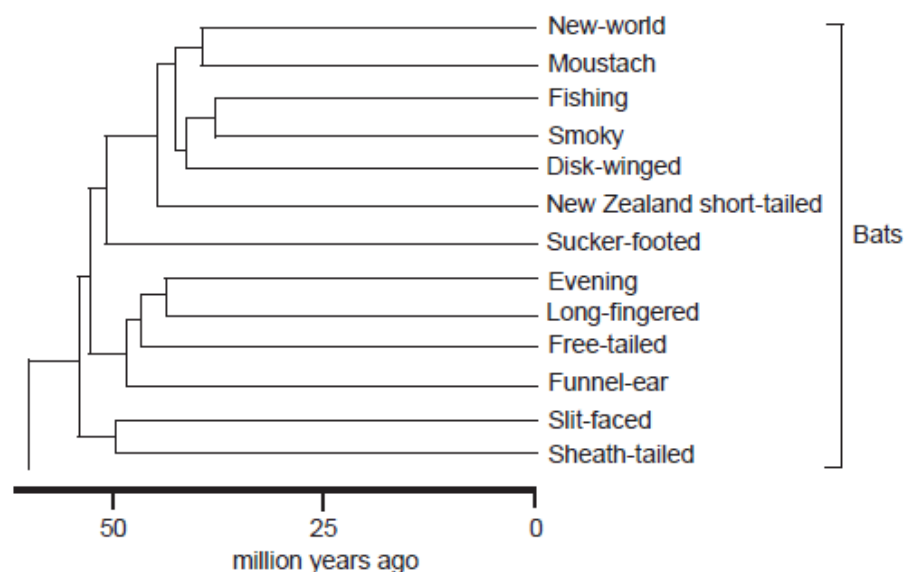
A

	filter pore size	growth of colonies on minimal medium
A	5µm 0.1µm	no yes
B	5µm 0.1µm	no no
C	0.45µm 0.02µm	no no
D	0.45µm 0.02µm	yes no

- 31 The mussel *Mytilus edulis* thrives in saline habitats, in both the highly salty seawater of tidal zones and the less salty estuaries. This result in two kinds of populations: one adapted to the higher salt concentrations of the tidal zone, and one adapted to the lower salt concentrations of the estuary. It has been found that the more salt-tolerant populations have high frequencies of an allele that produces an enzyme involved in maintaining osmotic equilibrium. Conversely, estuarine mussels having the same enzyme seem to be disfavoured and have a much higher death rate than mussels without the allele. Adult estuarine populations do have lower frequencies of this allele. Each spring, large numbers of larvae from the salty habitats pour into the estuaries.

The invasion of the seawater larvae would be expected to facilitate change in the genetic structure of the estuarine population by a process called

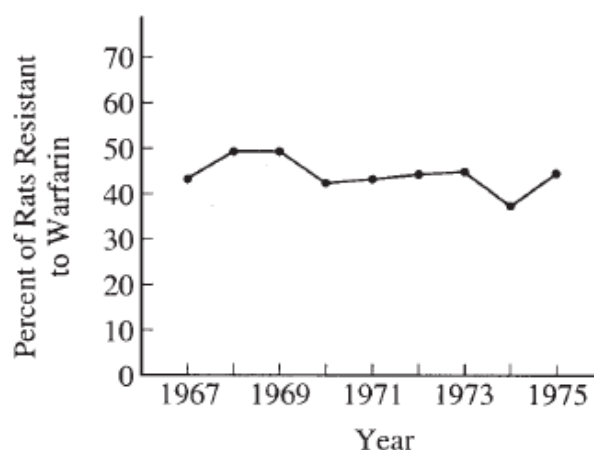
- A directional selection.
 - B genetic drift.
 - C gene flow.
 - D non-random mating.
- 32 Part of the phylogenetic tree for bats is shown below.



The most recent divergence of two bat species is the

- A smoky and fishing bats.
- B evening and long-fingered bats.
- C slit-faced and sheath-tailed bats.
- D sucker-footed and New Zealand short-tailed bats.

- 33 The Norway rat (*Rattus norvegicus*), a widespread pest, was controlled for about a decade by the anticoagulant warfarin. This chemical substance, placed in food pellets, is absorbed by the intestinal tract and inhibits clotting of blood. After a population decline for about 10 years, rat populations increased and stabilized. In one European population, as illustrated in the graph below, the percentage of rats resistant to warfarin has remained fairly stable over a number of years.



Resistance to warfarin is governed by a dominant autosomal gene, *R*. More than 15% of the resistant animals are heterozygous at this locus (*Rr*). The table below indicates the response to warfarin and relative reproductive fitness of individuals that are homozygous or heterozygous for the dominant gene (*R*). The *RR* individuals have a 20-fold increase in vitamin K requirement over *rr* individuals.

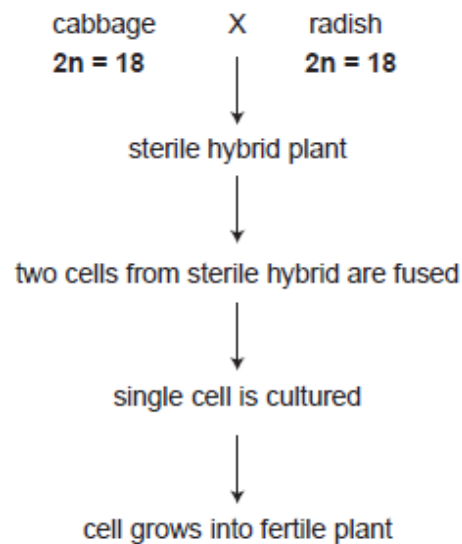
Relative fitness* at the R locus in the Norway rat in the presence of warfarin			
	<i>rr</i>	<i>Rr</i>	<i>RR</i>
Response to warfarin	Susceptible	Resistant	Resistant
Vitamin K dependence	No	Intermediate	Yes
Relative fitness	0.68	1.00	0.37

*Fitness is a measure of the reproductive success of a particular genotype. The highest fitness is 1.00.

There is a substantial number of heterozygotes in the population. Which of the following is the best explanation for this observation?

- A Heterozygotes (*Rr*) have a selective advantage over both homozygotes (*RR* and *rr*).
- B The gene for susceptibility (*r*) is increasing rapidly each generation.
- C Dominant homozygotes (*RR*) enjoy a reproductive advantage over heterozygotes.
- D Dominant homozygotes (*RR*) produce more offspring each generation than either heterozygotes (*Rr*) or recessive homozygotes (*rr*).

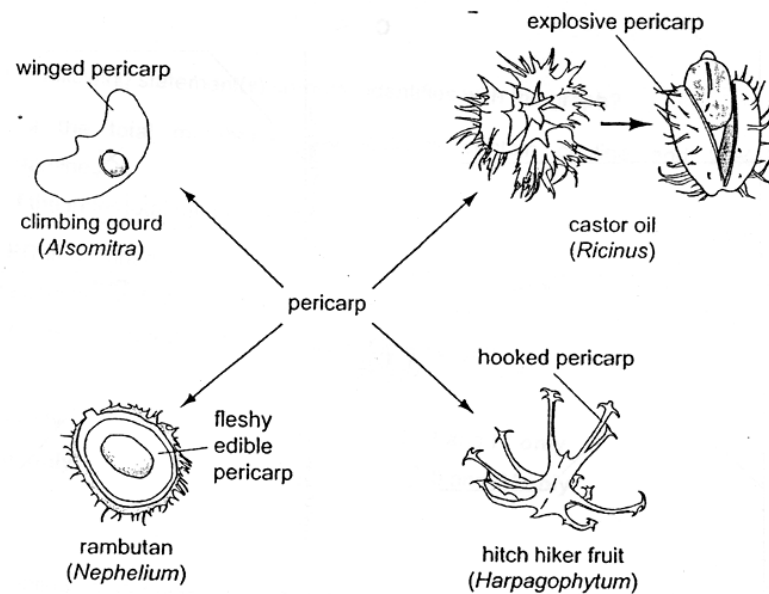
- 34 Cabbage, *Brassica oleracea*, and radish, *Raphanus sativus*, are both members of the Brassicaceae family. The cells of each type of plant have 18 chromosomes. When these plants are crossed, a sterile hybrid is produced. Two cells from a sterile hybrid plant are fused and the single cell that is formed is cultured. This is summarised in the following diagram.



From this information, it is reasonable to conclude that

- A cabbage and radish have identical chromosomes.
- B the fused cell has 18 pairs of homologous chromosomes.
- C the sterile hybrid has nine pairs of homologous chromosomes.
- D gametes from the fertile plant will be identical with those of the cabbage or the radish.

- 35 The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.



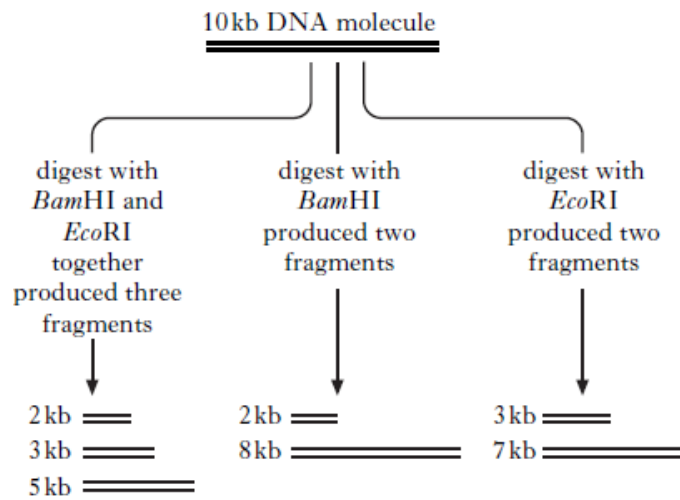
What do these examples illustrate?

- A The adaptive radiation of analogous structures showing divergent evolution.
 - B The adaptive radiation of analogous structures showing convergent evolution.
 - C The adaptive radiation of homologous structures showing divergent evolution.
 - D The adaptive radiation of homologous structures showing convergent evolution.
- 36 A soil bacterium (*Agrobacterium tumefaciens*) infects roses and fruit trees, stunting their growth. A similar bacterium (*A. radiobacter*) was genetically modified to include a plasmid gene coding for an antibiotic lethal to *A. tumefaciens*. *A. radiobacter* has a gene giving resistance to this antibiotic. A transfer gene located on the plasmid enables insertion of the modified gene into *A. radiobacter*.

Suspensions of genetically modified bacteria are applied to the soil around plants. For this treatment to be successful, genetic modification of *A. radiobacter* would need to include

- A removal of all plasmids.
- B removal of the antibiotic gene.
- C destruction of the antibiotic resistance gene.
- D removal of the plasmid transfer gene.

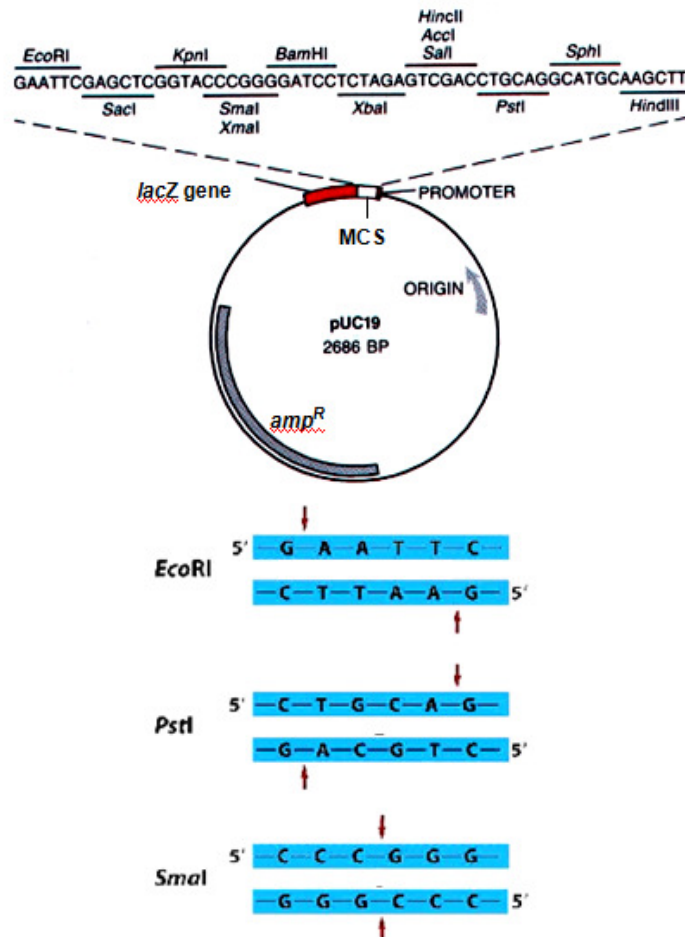
- 37 A piece of DNA was digested using the restriction enzymes *Bam*HI and *Eco*RI. The results are shown below.



Which of the following restriction maps can be drawn from these results?

- A**
- B**
- C**
- D**

- 38** The figure shows the plasmid map of pUC19, which includes the positions of the origin of replication (ORI), multiple cloning sites (MCS) and two genes - *amp^R* and *lacZ*. pUC19 is used for the production of human anti-thrombin III (hATIII) in the bacterium *E. coli*.



Which of the following can be concluded from the figure?

I	Any restriction enzyme whose restriction site is found within the MCS can be used to cleave pUC19.
II	pUC19 cleaved with <i>EcoRI</i> can anneal to hATIII cDNA cleaved with <i>PstI</i> as both restriction enzymes generate single-stranded overhangs of the same length.
III	The use of <i>SmaI</i> will require an additional step of ligating adaptors to the ends of pUC19 and hATIII cDNA.
IV	Colonies of <i>E. coli</i> transformed with recombinant pUC19 will survive on agar plates containing ampicillin and X-gal, giving a blue appearance.

- A** I and III only
- B** I and IV only
- C** II and III only
- D** All of the above

- 39** Pollen or anther culture involves growing haploid plantlets from the male gametes of the plant.

Embryo culture involves the removal of the embryo after successful fertilization of an ovule with pollen and the subsequent culture of the embryo to form a plantlet.

Which of the following is a similarity between the two plant tissue culture methods mentioned above?

- A** Resultant plants will be sterile.
- B** Homozygous plants can be produced in large numbers from calli.
- C** Roots are the first part to be produced via differentiation.
- D** Removal of the cell wall to form protoplasts may be done to facilitate the process of genetic modification.
- 40** In plant tissue culture, there are 5 main stages involved, starting from getting the explant to the acclimatisation of the plant.

Which of the following plant growth regulator ratios are used in the establishment of plantlets?

	relative amount of cytokinins	relative amount of auxins	ratio of plant growth regulator concentration
I	low	low	optimum ratio dependant on the type of plant
II	low	intermediate	intermediate auxin: cytokinin
III	high	low	high cytokinin:auxin
IV	low	high	high auxin:cytokinin

- A** I and II only
- B** I and IV only
- C** II and III only
- D** III and IV only

END OF PAPER